

# MTL106: Probability and Stochastic Processes

## Practice Sheet 2

1. A factory produces electronic gadgets. 95% of gadgets from Machine A are of high quality, and 90% of gadgets from Machine B are of high quality. Suppose 70% of the gadgets are produced by Machine A and 30% by Machine B. If a gadget is found to be of high quality, what is the probability that it was produced by Machine A?
2. Take four identical marbles. On the first, write  $A_1A_2A_3$ . On each of the three others, write  $A_1$ ,  $A_2$ , and  $A_3$ , respectively. Put the four marbles in an urn and draw one at random. Assume  $E_i$  denotes the event that the symbol  $A_i$  appears on the drawn marble. What can you say about the independence of  $E_1, E_2, E_3$  and pairwise independence of  $E_1, E_2, E_3$ ?
3. Let  $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$  and  $\mathcal{F} = \mathcal{P}(\Omega)$  (the power set). Define a mapping  $X : \Omega \rightarrow \mathbb{R}$  as follows:

$$X(\omega_1) = 1, \quad X(\omega_2) = 1, \quad X(\omega_3) = 2, \quad X(\omega_4) = 0$$

Determine the inverse image set  $X^{-1}(-\infty, x]$  for the following values of  $x$ :

- (a)  $x = 0.5$
- (b)  $x = 1.5$
- (c)  $x = 2.5$

Conclude whether  $X$  is a random variable on  $(\Omega, \mathcal{F})$ .

4. Let  $S$  be the set of all binary strings of length 50, where each entry is either 0 or 1. Assume that each string in  $S$  is equally likely. Define the random variable  $X$  as the number of 1s in a randomly selected string from  $S$ .
  - (a) Write down the sample space  $S$  and determine its cardinality.
  - (b) Specify the support (range) of the random variable  $X$ .
  - (c) Find the probability mass function  $P_X(i) = P(X = i)$ , for  $i = 0, 1, 2, \dots, 50$ .
  - (d) Using the result obtained in part (c), compute  $P(X = 27)$ .
5. Consider a fair coin tossed twice. Let:  $E_1$ : Head on the first toss,  $E_2$ : Head on the second toss and  $E_3$ : Exactly one Head appears. Show that these events are pairwise independent but not mutually independent.

6. Consider the function  $F : \mathbb{R} \rightarrow \mathbb{R}$  defined as:

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x^2}{4} & 0 \leq x < 1 \\ \frac{1}{2} & 1 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$$

- (a) Verify whether  $F(x)$  is a valid cumulative distribution function by checking the properties:
- (b) Identify the points of discontinuity. Using the result  $P(X = x_j) = F(x_j) - F(x_{j-1})$  (from the limit from the left), determine the probability mass at these jump points.
- (c) Calculate  $P(0.5 < X \leq 1.5)$ .

7. A random variable  $X$  has the following probability distribution:

$$P(X = x) = \begin{cases} k \cdot (0.5)^x & x = 1, 2, 3 \\ k \cdot x & x = 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the value of  $k$ .
- (b) Construct the CDF,  $F(x)$  and sketch its graph.

8. A certain river floods every year. Suppose that the low-water mark is set at 1 and the high-water mark  $Y$  has cumulative distribution function

$$F_Y(y) = \begin{cases} 0, & y < 1, \\ 1 - \frac{1}{y^2}, & y \geq 1. \end{cases}$$

- (a) Verify that  $F_Y(y)$  is a valid cdf.
- (b) Find the probability density function  $f_Y(y)$  of  $Y$ .
- (c) Suppose the low water mark is reset at 0 and measurements are taken in units that are  $1/k$  times the original unit. If

$$X = k(Y - 1),$$

find the cumulative distribution function of  $X$ .